

Cyclic Lineage Dynamics and Rank-Five First-Exit Frontiers

An All- n Classification for the Canonical Single-Defect Circular Family

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This paper (Paper XXIX of the RIME program) develops an intrinsic raw-geometry theory for rank-five packet frontiers in single-defect circular automata. It is independently scoped from the typed transfer and recursive-return contracts of Paper XXVIII. Those contracts motivate an open bridge but are not premises of the theorems proved here.

Abstract

Problem. A rank- $(n - 1)$ defect merges one pair of coordinates, but a rank-five first exit depends on more than the five-point support or the pair that eventually meets the kernel. The three surviving source packets retain their own placement, and every earlier defect step must remain injective on all five packets.

Approach. We encode raw packet states in defect-anchored cyclic coordinates, factor each defect as one marked collapse followed by a permutation, and lift the plateau graph to source-packet lineages. For a candidate fused pair, quotienting internal pair and spectator labels gives an exact $2 + 3$ spectator-safe configuration system. In the canonical family

$$d_n(0) = d_n(1) = 0, \quad d_n(q) = q - 1 \quad (2 \leq q \leq n - 1),$$

this system admits a missing-image section, a universal cyclic return, and a constructive hole-slide normal form.

Results. For every $n \geq 5$, a candidate source pair can become the first fused pair if and only if its two lineages are consecutive in the positive cyclic order of the five source packets. Every cyclic-order-preserving placement of the five lineages is plateau-reachable. Consequently, for a reachable fused pair $F = \{f_0, f_1\}$ with remaining source order (r_1, r_2, r_3) , the survivor spectrum is exactly

$$1 \leq s(r_1) < s(r_2) < s(r_3) \leq n - 2.$$

The complete source-addressed raw first-exit frontier therefore has

$$5 \binom{n-2}{3}$$

endpoints. Its incidence type depends on source packets only through their labels and cyclic order, not their masses or internal atom composition.

Boundary. The classification is a raw first-exit theorem. It does not establish transfer membership, entry authorization, typed handoff, projectability, recursive return, credit settlement, or a reset bound. The extension from raw incidence to the declared projectable transfer fibers of Paper XXVIII remains open.

Keywords. synchronizing automata; circular automata; rank defect; packet lineage; first-exit frontier; spectator-safe reachability; cyclic order

Notation Table

Symbol	Meaning
$Q = \mathbb{Z}/n\mathbb{Z}$	labelled cyclic state set, with $n \geq 5$
$p(q) = q + 1$	positive labelled cycle on Q
$d : Q \rightarrow Q$	labelled rank- $(n - 1)$ defect with binary kernel
$K_d = \{k_0, k_1\}, c_d, m_d$	kernel, collision image, and missing image of d
$P = (T, \Pi)$	raw packet state with occupied support T and source-addressed blocks Π
$\mathcal{A}$	fixed labelled atom carrier partitioned by the packets
$P_0 = (T_0, \Pi_0)$	fixed raw rank-five source
$\lambda : T_0 \hookrightarrow Q$	current placement of the five source-packet lineages
$\mathcal{L}_5(d; P_0)$	plateau-reachable lineage injections
$\mathfrak{B}_d(P_0)$	boundary-incidence spectrum $\{d \circ \lambda\}$
$\mathfrak{F}_d(P_0)$	reachable fused source-pair spectrum
$\mathfrak{S}_d(P_0; F)$	survivor-placement spectrum over F
$\text{Conf}_{2,3}(Q)$	marked pair plus spectator-triple configurations
(A, H)	marked-pair and hole presentation of a $2 + 3$ configuration
$\Omega(A, H)$	cyclic order of the five non-hole tokens
d_n	canonical defect $0, 1 \mapsto 0$ and $q \mapsto q - 1$ for $q \geq 2$

1 Introduction

A synchronizing automaton acts simultaneously on states, subsets, and fibers of source labels. Those levels coincide only in special circumstances. A five-point support may reveal where a rank drop occurs, while failing to record which source packets fuse or where the surviving packets land. For a first-exit argument, that omitted incidence is part of the result.

This paper isolates the intrinsic geometry of the first rank drop in a single-defect circular automaton. Let p be a labelled cycle and let d have rank $n - 1$ with one binary kernel. Starting from five source-addressed packets, we follow words for which every intermediate step preserves rank five. The next d -step is then strict and produces a rank-four endpoint. Our question is:

Which source-addressed rank-four endpoints can occur at that first exit?

The answer is complete for the canonical family d_n . Its short form is as follows.

Main Theorem. Fix $n \geq 5$ and a raw rank-five source whose packet lineages have positive cyclic order $(q_0, q_1, q_2, q_3, q_4)$. Under the canonical defect d_n :

1. the possible fused source pairs are exactly the five consecutive pairs $\{q_i, q_{i+1}\}$;
2. after writing a fused pair as (f_0, f_1) in positive order and the three survivors as (r_1, r_2, r_3) , its possible output placements are exactly the increasing injections

$$1 \leq s(r_1) < s(r_2) < s(r_3) \leq n - 2;$$

3. the complete raw frontier contains $5\binom{n-2}{3}$ endpoints.

The proof requires more than ordinary pair hitting. The three other packets act as spectators: every internal d -step must be injective on the entire five-point support. We therefore pass through the exact quotient

$$\text{five labelled lineages} \twoheadrightarrow \text{Conf}_{2,3}(Q),$$

which forgets identities inside the candidate pair and inside the spectator triple while retaining the complete collision guard. In token/hole form, the canonical dynamics preserve the cyclic

order of the five non-hole tokens. Conversely, an explicit section return slides holes across adjacent tokens. This turns token-order adjacency into a necessary and sufficient, path-free criterion for fused-pair reachability.

The paper contributes five linked pieces.

1. A defect-anchored coordinate system faithfully encodes raw packet states and their first-exit frontiers.
2. A matched control proves that kernel, root, missing image, support gaps, and source blocks do not determine the source-addressed frontier for a general labelled defect.
3. The factorization $d = \rho_d \circ e_K$ separates one marked collapse from the off-kernel permutation transport, and the packet-lineage lift records exactly the missing incidence.
4. The $2 + 3$ spectator-safe quotient is exact for fused-pair reachability, whereas the further pair-only quotient is not.
5. For d_n , a missing-image section, cyclic return symmetry, and hole-slide normal form give the fused-pair criterion and the complete survivor spectrum above.

The result is deliberately below the typed transfer layer. Paper XXVIII established finite projectable transfer fibers and left their all- n supply open [3]. The present theorem classifies raw first exits without assuming that every raw endpoint belongs to a declared transfer fiber or carries a projection witness. Section 7 records that bridge as an open problem rather than importing the missing semantics into the raw object.

1.1 Related Work and Novelty Boundary

For the canonical family, let b be the standard Cerny defect letter, $b(n-1) = 0$ and $b(q) = q$ for $0 \leq q \leq n-2$ [2, 8]. Then

$$b = d_n \circ p, \quad d_n = b \circ p^{-1}. \quad (1.1)$$

Since $p^{-1} = p^{n-1}$, the alphabets $\{p, d_n\}$ and $\{p, b\}$ generate the same transition monoid. The canonical theorem therefore concerns a finer observable inside the standard Cerny transition system, not a new monoid.

The Cerny conjecture asks for a quadratic reset bound for synchronizing automata [2, 10, 11]. Circular automata admit stronger uniform results under hypotheses different from those used here [6]. Almost-group automata, consisting of permutation letters and a letter with one nontrivial kernel class, provide a nearby structural setting [1]. This paper does not prove a reset bound or classify general almost-group automata. It fixes one cyclic permutation, one labelled binary-kernel defect, and one rank-five first-exit problem.

Eppstein's monotonic setting studies automata whose letters preserve a fixed cyclic order and gives algorithms and bounds for reset sequences [7]. Subset and complete reachability ask which unlabelled subsets occur as images of the full state set. Don proves complete subset reachability for an aperiodically 1-contracting class and develops the circular special case [5]. Gonze and Jungers exhibit obstructions and long witnesses for subset reachability and refine the structural theory of completely reachable automata [9]. Ferens and Szykula give a quadratic-time recognition algorithm and reaching-threshold bounds for completely reachable automata, a class that includes the Cerny automata [8]. These results establish cyclic-order, support, or subset reachability statements. They do not by themselves classify paths that must remain rank-five at every proper prefix while retaining source-packet lineages, the three-spectator collision guard, and the complete terminal incidence map. Those plateau-constrained, source-addressed observables are the additional claims of this paper.

Within the RIME synchronizing-automata line, Paper XXIII studies pair hitting, marked-kernel corridors, and Schreier waiting [4]. Its pair-hitting language motivates the present comparison, but ordinary two-point reachability is too coarse here: three spectator lineages determine whether an internal defect step is legal. Paper XXVIII studies finite typed transfer and return relations [3]. The novelty here is an all- n raw theorem: the exact spectator-safe quotient, the canonical token-order normal form, and the complete source-addressed rank-five frontier classification. No theorem of the cited papers is restated as this classification.

2 Raw Packet Dynamics and First Exit

Let $Q = \mathbb{Z}/n\mathbb{Z}$, $n \geq 5$, and let $p(q) = q + 1$. A labelled defect $d : Q \rightarrow Q$ has rank $n - 1$. Hence it has one non-singleton fiber

$$K_d = \{k_0, k_1\}, \quad d(k_0) = d(k_1) = c_d,$$

and one missing image $m_d \notin \text{im}(d)$. All other fibers are singletons.

A raw rank- r packet state is a pair $P = (T, \Pi)$ with $T \subseteq Q$, $|T| = r$, and nonempty, pairwise disjoint atom blocks $\Pi(q)$, $q \in T$, whose union is a fixed labelled atom carrier $\mathcal{A}$ with $|\mathcal{A}| = n$. A letter acts by push-forward:

$$a_*P = \left(aT, b \mapsto \bigcup_{\substack{q \in T \\ a(q)=b}} \Pi(q) \right), \quad a \in \{p, d\}. \quad (2.1)$$

Thus letters move packet coordinates and union packets with a common image; they do not rename atoms. We suppress the subscript $*$ when no ambiguity is possible.

Fix a raw rank-five source $P_0 = (T_0, \Pi_0)$. A word $w = a_1 \cdots a_k$ is a **plateau word** from P_0 if every prefix acts with rank five. Words act from left to right, so the corresponding coordinate map after the prefix $a_1 \cdots a_i$ is $a_i \circ \cdots \circ a_1$. A first-exit endpoint has the form

$$d(wP_0),$$

where w is a plateau word and the final d drops the rank from five to four. The complete raw frontier is

$$\text{Fr}_{5 \rightarrow 4, d}^{\text{raw}}(P_0) = \{d(wP_0) : w \text{ is plateau-admissible and the final } d \text{ is strict}\}. \quad (2.2)$$

Because K_d is the unique non-singleton fiber,

$$|dT| = |T| - 1 \iff K_d \subseteq T. \quad (2.3)$$

Hence a d -step is plateau-admissible exactly when the current support does not contain both kernel points, and the terminal d -step is strict exactly when it does.

The canonical family studied in Sections 5 and 6 is

$$d_n(0) = d_n(1) = 0, \quad d_n(q) = q - 1 \quad (2 \leq q \leq n - 1). \quad (2.4)$$

It has kernel $K = \{0, 1\}$, collision root $c = 0$, and missing image $m = n - 1$.

3 Faithful Coordinates and a Coarse Obstruction

3.1 Defect-anchored cyclic coordinates

For a general labelled pair (p, d) , use the collision root to identify Q with $\mathbb{Z}/n\mathbb{Z}$ by

$$\iota_d(p^j(c_d)) = j.$$

Retain the full labelled map

$$\delta_d = \iota_d \circ d \circ \iota_d^{-1}.$$

Let $P = (T, \Pi)$ have occupied indices $j_0 < \dots < j_{r-1}$. Put

$$B_i = \Pi(\iota_d^{-1}(j_i)), \quad g_i = j_{i+1} - j_i \quad (i < r-1), \quad g_{r-1} = n + j_0 - j_{r-1},$$

and define

$$\mathfrak{G}_r(P) = (j_0, (g_0, \dots, g_{r-1}), (B_0, \dots, B_{r-1})). \quad (3.1)$$

An admissible encoded state has $g_i > 0$, $\sum_i g_i = n$, $0 \leq j_0 < g_{r-1}$, and ordered nonempty blocks B_i partitioning $\mathcal{A}$. The phase condition reconstructs ordinary sorted indices $j_i = j_0 + \sum_{h < i} g_h < n$; gaps alone would leave a cyclic reindexing ambiguity.

Encoded moves act on the pairs (j, B) . The p -move sends them to $(j+1 \bmod n, B)$; the d -move sends them to $(\delta_d(j), B)$ and unions blocks with equal images. In both cases, sort the resulting pairs by their new indices, carry the blocks through the same reindexing, and recompute phase and gaps. Denote these operations by $\hat{p}$ and $\hat{d}$.

Proposition 3.1 (faithful raw encoding).

For $1 \leq r \leq n$, $\mathfrak{G}_r$ is a bijection between raw rank- r packet states and the admissible encoded states above. If $a \in \{p, d\}$ sends P from rank r to rank r' , then

$$\mathfrak{G}_{r'}(aP) = \hat{a}(\mathfrak{G}_r(P)). \quad (3.2)$$

Consequently, $\mathfrak{G}_4$ maps the raw first-exit frontier bijectively to the frontier obtained by the encoded transitions and the same rank-profile rule.

Proof. From an admissible tuple, reconstruct $j_i = j_0 + \sum_{h < i} g_h$. Positivity orders the indices strictly, and $j_{r-1} = j_0 + n - g_{r-1} < n$ by the phase condition. Place B_i at $\iota_d^{-1}(j_i)$. This decodes the tuple, while encoding the decoded state returns it. Conversely, every sorted support satisfies $g_{r-1} = n + j_0 - j_{r-1} > j_0$, so its encoding is admissible.

The identities $\iota_d(p(q)) = \iota_d(q) + 1 \bmod n$ and $\iota_d(d(q)) = \delta_d(\iota_d(q))$ give (3.2), including the required block reindexing when an occupied coordinate crosses $n-1 \rightarrow 0$. Equation (2.3) gives the rank test. Induction over a plateau word preserves the full raw state and the rank of every prefix; the terminal step is strict on one side exactly when it is strict on the other. This proves frontier faithfulness.

The full map δ_d is essential in this statement. Kernel, collision root, and missing image do not determine off-kernel transport.

3.2 A quotient through which the frontier does not descend

Proposition 3.2 (coarse frontier non-descent).

Let $Q = \mathcal{A} = \mathbb{Z}/5\mathbb{Z}$, let P_* place the singleton packet $\{q\}$ at q , and define

$$d = (0, 0, 1, 2, 3), \quad d' = (0, 0, 2, 1, 3), \quad (3.3)$$

where the lists give images at inputs 0, 1, 2, 3, 4. The two defects have the same cycle, kernel $\{0, 1\}$, collision root 0, missing image 4, and initial gap/block data, but

$$\text{Fr}_{5 \rightarrow 4, d}^{\text{raw}}(P_*) \cap \text{Fr}_{5 \rightarrow 4, d'}^{\text{raw}}(P_*) = \emptyset. \quad (3.4)$$

Proof. Every rank-five support is all of Q , so no d -letter is allowed on the plateau. A candidate endpoint is therefore $d(p^t P_*)$ or $d'(p^t P_*)$, with t modulo five. In either endpoint the collision-root packet is $\{-t, 1 - t\}$. These five unordered pairs are distinct, so equality of endpoints would force the same t . For that t , however, the packet at coordinate 1 is $\{2 - t\}$ for d and $\{3 - t\}$ for d' . The endpoints are unequal.

Thus the source-addressed frontier is not a function of the cycle, kernel, root, missing image, and initial gap/block data. The example does not prove that the entire defect table is irreducible; it rules out this particular coarse quotient. It also proves a sharper warning used later: output support and the fused source pair need not determine the surviving packet placement.

3.3 One marked collapse and one permutation

Choose an orientation $K_d = \{k_0, k_1\}$ and define

$$e_K(k_1) = k_0, \quad e_K(q) = q \quad (q \neq k_1).$$

Proposition 3.3 (permutation-plus-collapse factorization).

There is a unique permutation ρ_d with

$$\rho_d(q) = d(q) \quad (q \neq k_1), \quad \rho_d(k_1) = m_d, \quad (3.5)$$

and

$$\boxed{d = \rho_d \circ e_K}. \quad (3.6)$$

For every raw packet state,

$$d_* P = (\rho_d)_* ((e_K)_* P). \quad (3.7)$$

The only possible union in $(e_K)_* P$ is the packet pair occupying K_d .

Proof. The restriction

$$d|_{Q \setminus \{k_1\}} : Q \setminus \{k_1\} \xrightarrow{\sim} \text{im}(d)$$

is bijective: it is injective because the only collision of d is K_d , and its domain and codomain both have $n-1$ elements. Assigning the remaining input k_1 to the remaining output m_d gives the unique permutation. Equation (3.6) follows at every input, and push-forward under a composite map gives (3.7).

The rank-five plateau is therefore a system of five labelled particles under permutation transport and one marked substitution. A strict exit occurs when both marked kernel coordinates are occupied. The next section records the particle lineages that support data alone forgets.

4 Packet Lineages and the Exact Spectator Quotient

4.1 The lineage lift and boundary incidence

Fix $P_0 = (T_0, \Pi_0)$. Before the first strict exit there is no packet fusion or splitting. Every current packet is therefore one unique source packet. Record this by an injection

$$\lambda : T_0 \hookrightarrow Q,$$

where the source block $\Pi_0(q)$ currently occupies $\lambda(q)$. Its image is the current support. A letter $a \in \{p, d\}$ is a legal plateau step exactly when a is injective on $\text{im } \lambda$, and then

$$\lambda \xrightarrow{a} a \circ \lambda. \quad (4.1)$$

Let $\mathcal{L}_5(d; P_0)$ be the least set of injections containing the inclusion $T_0 \hookrightarrow Q$ and closed under (4.1) whenever the step is legal. Let

$$\mathcal{L}_5^\partial(d; P_0) = \{\lambda \in \mathcal{L}_5(d; P_0) : K_d \subseteq \text{im } \lambda\}. \quad (4.2)$$

Theorem 4.1 (packet-lineage lift).

The map

$$\lambda \mapsto (\text{im } \lambda, q \mapsto \Pi_0(\lambda^{-1}(q))) \quad (4.3)$$

is a bijection from $\mathcal{L}_5(d; P_0)$ to the raw rank-five packet states plateau-reachable from P_0 . It intertwines every legal p - or d -step. For $\lambda \in \mathcal{L}_5^\partial(d; P_0)$, put

$$\beta = d \circ \lambda : T_0 \rightarrow d(\text{im } \lambda). \quad (4.4)$$

The corresponding strict-exit endpoint is

$$\Theta_{P_0}(\beta) = \left(\text{im } \beta, b \mapsto \bigcup_{\substack{q \in T_0 \\ \beta(q)=b}} \Pi_0(q) \right). \quad (4.5)$$

Moreover, Θ_{P_0} is a bijection from the boundary-incidence spectrum

$$\mathfrak{B}_d(P_0) = \{d \circ \lambda : \lambda \in \mathcal{L}_5^\partial(d; P_0)\} \quad (4.6)$$

onto $\text{Fr}_{5 \rightarrow 4, d}^{\text{raw}}(P_0)$.

Proof. Initially (4.3) is P_0 . If a is injective on the current support, push-forward moves the block at $\lambda(q)$ to $a(\lambda(q))$ without union or splitting. This is exactly the state decoded from $a \circ \lambda$. Induction gives surjectivity onto the plateau-reachable states. Since the source blocks are nonempty and disjoint, each current block identifies its unique source block, so the state recovers λ .

At a terminal support containing K_d , the only non-singleton fiber of d is the kernel pair. Formula (4.5) is therefore precisely the strict push-forward. Every source block occurs in exactly one output block, whose coordinate recovers $\beta(q)$. Thus Θ_{P_0} is both surjective and injective.

Every $\beta \in \mathfrak{B}_d(P_0)$ has a unique decomposition

$$F = \beta^{-1}(c_d), \quad s = \beta|_{T_0 \setminus F}. \quad (4.7)$$

Here F is the fused source pair and s is the placement of the three surviving lineages. Define

$$\mathfrak{F}_d(P_0) = \{\beta^{-1}(c_d) : \beta \in \mathfrak{B}_d(P_0)\} \quad (4.8)$$

and, for $F \in \mathfrak{F}_d(P_0)$,

$$\mathfrak{S}_d(P_0; F) = \{\beta|_{T_0 \setminus F} : \beta \in \mathfrak{B}_d(P_0), \beta^{-1}(c_d) = F\}. \quad (4.9)$$

Then

$$\mathfrak{B}_d(P_0) \cong \bigsqcup_{F \in \mathfrak{F}_d(P_0)} \{F\} \times \mathfrak{S}_d(P_0; F). \quad (4.10)$$

Equation (4.10) is a decomposition of attained incidences, not a claim that all formal pairs (F, s) occur. Proposition 3.2 shows why both coordinates matter: its two defects have the same support and fused source pair but different survivor placements.

If source and current supports are sorted in defect-anchored coordinates, the injection λ is represented by a permutation in S_5 . Each legal plateau edge induces an S_5 -valued reindexing $\alpha_{a,T}$. Along a composable path its ordered product is the total lineage permutation. This edge cocycle records plateau transport; the terminal map $d \circ \lambda$ is additional boundary data. In particular, Proposition 3.2 has the same plateau cocycle on both sides and differs only at the terminal survivor map.

4.2 Pair steering with three spectators

Fix a candidate pair $F \subset T_0$, $|F| = 2$, and put $R_0 = T_0 \setminus F$. Define

$$\text{Conf}_{2,3}(Q) = \{(A, R) : |A| = 2, |R| = 3, A \cap R = \emptyset\}. \quad (4.11)$$

The p -transition is always defined and sends (A, R) to (pA, pR) . The d -transition is defined precisely when

$$K_d \not\subseteq A \cup R, \quad (4.12)$$

and then sends (A, R) to (dA, dR) . The target set is

$$\mathcal{K}_d = \{(K_d, R) : |R| = 3, R \cap K_d = \emptyset\}. \quad (4.13)$$

The guard (4.12) requires the defect to remain injective on both the marked pair and all three spectators. This is the information ordinary pair hitting omits.

Theorem 4.2 (exact spectator-safe quotient).

The map

$$q_F(\lambda) = (\lambda(F), \lambda(R_0)) \quad (4.14)$$

is the quotient of the full lineage state space by independent relabelling inside F and inside R_0 . It intertwines the partial p, d transitions, maps reachable lineage states onto the reachable subgraph of $\text{Conf}_{2,3}(Q)$, and satisfies

$$\boxed{F \in \mathfrak{F}_d(P_0) \iff (F, R_0) \rightsquigarrow \mathcal{K}_d \text{ in } \text{Conf}_{2,3}(Q)}. \quad (4.15)$$

Proof. Every (A, R) has a preimage under q_F : combine arbitrary bijections $F \rightarrow A$ and $R_0 \rightarrow R$. Two injections have the same image exactly when their restrictions differ by precomposition with an element of $S_2 \times S_3$. If a is injective on the five-point support, then

$$q_F(a \circ \lambda) = (a\lambda(F), a\lambda(R_0)) = a q_F(\lambda).$$

For d , injectivity is exactly (4.12); for p it is automatic. Hence every lineage path projects. Conversely, applying the labels of a quotient path to the initial inclusion produces an injection at every stage because each quotient edge satisfies the same guard. The final state lies in $\mathcal{K}_d$ exactly when the lifted injection sends F onto K_d . Equations (4.6) and (4.8) now give (4.15).

The quotient has

$$|\text{Conf}_{2,3}(Q)| = \binom{n}{2} \binom{n-2}{3} = 10 \binom{n}{5} \quad (4.16)$$

states. It forgets only internal $S_2 \times S_3$ labels. The further map $(A, R) \mapsto A$ is not an exact quotient for target reachability: the spectator triple determines whether d is enabled. Thus

$$\text{full lineage lift} \twoheadrightarrow \text{Conf}_{2,3}(Q) \not\rightarrow \binom{Q}{2} \quad (4.17)$$

is the relevant descent boundary.

4.3 Token and hole form

Color the coordinates by P, S, and H, with multiplicities 2, 3, and $n - 5$. The marked pair is $A = c^{-1}(\text{P})$, the spectators are $R = c^{-1}(\text{S})$, and the holes are

$$H = Q \setminus (A \cup R) = c^{-1}(\text{H}).$$

Equivalently, retain (A, H) , since R is its complement. Under the factorization $d = \rho_d \circ e_K$, define

$$\bar{e}_K(H) = Q \setminus e_K(Q \setminus H).$$

The exact partial transitions become

$$p : (A, H) \mapsto (pA, pH),$$

$$d : (A, H) \mapsto (\rho_d(e_K(A)), \rho_d(\bar{e}_K(H))), \quad H \cap K_d \neq \emptyset. \quad (4.18)$$

The target condition is $A = K_d$. Formula (4.18) turns the spectator guard into a hole-shield rule: every internal d -step requires at least one hole on the kernel, while the terminal configuration places the two marked tokens on both kernel points.

5 Canonical Spectator-Safe Reachability

From now on $d = d_n$ is the canonical defect (2.4), with $K = \{0, 1\}$ and $m = n - 1$.

5.1 The missing-image section

For a general rank- $(n - 1)$ defect, every guarded d -output omits m_d . This gives the section

$$\Sigma_{m_d} = \{(A, H) : m_d \in H\}. \quad (5.1)$$

For $t \in \mathbb{Z}/n\mathbb{Z}$, define the partial return

$$R_t = d \circ p^t, \quad \text{Dom}(R_t) = \{(A, H) \in \Sigma_{m_d} : p^t H \cap K_d \neq \emptyset\}. \quad (5.2)$$

Here R_t acts on the whole token/hole state. Every R_t returns to Σ_{m_d} .

Proposition 5.1 (canonical section symmetry).

Let $n \geq 6$ and identify $D = Q \setminus \{m\}$ with $\mathbb{Z}/(n - 1)\mathbb{Z}$. On $\Sigma_n := \Sigma_m$, the returns R_1 and R_2 are defined everywhere and

$$\boxed{R_1 = \text{id}}, \quad \boxed{R_2 = \tau}, \quad \tau(r) = r + 1 \pmod{n - 1} \quad (r \in D), \quad (5.3)$$

where R_2 fixes the missing hole m and rotates all colors on D .

Proof. A guarded defect output has support in $\text{im}(d) = Q \setminus \{m_d\}$, proving section entry. In the canonical section, m is a hole. Since $p(m) = 0$ and $p^2(m) = 1$, both returns are guarded. For every $r \in D$,

$$d_n(p(r)) = r, \quad d_n(p^2(r)) = r + 1 \pmod{n - 1}.$$

These identities hold simultaneously for every token and every extra hole, which proves (5.3).

Thus R_2 gives a universal C_{n-1} symmetry on the section. The quotient $\Sigma_n/\langle R_2 \rangle$ consists of cyclic necklaces on D with two P tokens, three S tokens, and $n - 6$ extra holes.

5.2 The token-order invariant

For any token/hole state, read colors cyclically around Q and delete every H . The resulting length-five cyclic word is the **hole-deleted token order** $\Omega(A, H)$. It has two possible types:

$$[PPSSS] \quad \text{or} \quad [PSPSS]. \quad (5.4)$$

Let $\text{Adj}_\Omega(A, H)$ be 1 in the first case and 0 in the second.

Lemma 5.2 (order preservation).

Every p -step preserves Ω . Every guarded canonical d_n -step also preserves Ω .

Proof. The cycle rotates the complete coloring. For a guarded defect step, at least one of $0, 1$ is a hole. If 0 is a hole, d_n is strictly order-preserving on the occupied subset of $\{1, \dots, n-1\}$. If 1 is a hole, it is strictly order-preserving on the occupied subset of $\{0, 2, \dots, n-1\}$. The case in which both are holes satisfies both descriptions. Deleting holes therefore gives the same cyclic token word.

Lemma 5.3 (hole slide).

Assume $n \geq 7$ and $(A, H) \in \Sigma_n$. In the necklace quotient, a local block HX , with $X \in \{P, S\}$, can be replaced by XH without changing the cyclic order of the other colors.

Proof. Use a power of the universal return R_2 to place the chosen block at coordinates $n-3, n-2$. The return $R_3 = d_n \circ p^3$ is guarded because the selected hole moves to kernel coordinate 0. If W is the remaining linear word on D , direct evaluation gives

$$WHX \xrightarrow{R_3} XHW \sim_{R_2} WXH. \quad (5.5)$$

The final equivalence is another power of R_2 . The selected hole crosses exactly the adjacent token X .

The lemma is constructive: extra holes can move through the fixed cyclic order of the five non-hole tokens. They change physical gaps but not Ω .

Labelled-lift consequence. The word constructed in Lemma 5.3 also acts on a fully labelled lineage state. After the same final R_2 phase alignment, it exchanges the selected hole with the selected adjacent lineage and preserves the cyclic order and identities of every other labelled lineage. Indeed, the construction selects a coordinate hole and applies the fixed coordinate maps R_2 and R_3 ; it never branches on whether the adjacent non-hole token has color P or S , nor on its lineage label. Thus forgetting lineage labels before the move gives exactly the color move of Lemma 5.3.

5.3 The path-free fused-pair criterion

Theorem 5.4 (all- n canonical spectator-safe classification).

For every $n \geq 5$ and every $(A, R) \in \text{Conf}_{2,3}(Q)$, with $H = Q \setminus (A \cup R)$,

$$\boxed{(A, R) \rightsquigarrow \mathcal{K}_{d_n} \iff \text{Adj}_\Omega(A, H) = 1.} \quad (5.6)$$

Equivalently, for a fixed raw source P_0 and $F \subset T_0$, $|F| = 2$,

$$\boxed{F \in \mathfrak{F}_{d_n}(P_0) \iff \text{the members of } F \text{ are consecutive in the cyclic order of } T_0.} \quad (5.7)$$

Proof. Lemma 5.2 makes Adj_Ω invariant along every spectator-safe path. At the target the marked pair is $K = \{0, 1\}$, so its tokens are adjacent. This proves necessity.

For sufficiency, first let $n = 5$. There are no holes, and a power of p sends any adjacent marked pair to K . Let $n \geq 6$. If the state is not in Σ_n , choose a hole, rotate it to 0, and apply d_n . This guarded step enters the section and preserves Ω .

When $n = 6$, there are no extra holes on D . Marked tokens adjacent in Ω are physically adjacent on the five-cycle D , and a power of R_2 sends them to K . Let $n \geq 7$. Choose the two consecutive marked tokens in Ω . If their physical gap contains holes, apply Lemma 5.3 to the last hole before the second marked token. Each slide removes one hole from that gap and introduces none. Repetition makes the marked tokens physically adjacent, after which a power of R_2 sends them to K . This proves (5.6).

At the initial lineage state, Ω is exactly the cyclic order of T_0 with the members of F marked. The exact quotient theorem (4.15) therefore turns (5.6) into (5.7).

The bit Adj_Ω is complete for this target-reachability predicate. We do not claim that Ω is a deterministic transition congruence, that equal Ω -states have equal shortest words, or that they are all mutually reachable. The stronger lineage statement needed for the full frontier is proved next.

6 Complete Canonical First-Exit Frontier

Give T_0 and Q their positive cyclic orders induced by p . An injection $\lambda : T_0 \hookrightarrow Q$ is **cyclic-order preserving** if it preserves the oriented cyclic order of all five source lineages.

6.1 Reachability of the complete order fiber

Theorem 6.1 (canonical plateau order-fiber transitivity).

For every $n \geq 5$ and every fixed raw rank-five source P_0 ,

$$\mathcal{L}_5(d_n; P_0) = \{\lambda : T_0 \hookrightarrow Q : \lambda \text{ preserves cyclic order}\}. \quad (6.1)$$

The partial plateau graph induced on this set is strongly connected.

Proof. Lemma 5.2 remains valid when the five source lineages are given distinct labels. Hence every plateau transition preserves their oriented cyclic order, proving the forward inclusion.

For $n = 5$, every cyclic-order-preserving bijection of the five-cycle is a power of p . Assume $n \geq 6$. Every five-lineage injection has a hole. Rotating a hole to 0 and applying d_n gives a guarded entry to the missing-image section without changing lineage order.

Fix the positive source order $q_0, \dots, q_4$. For a section injection, let g_i be the number of extra holes on $D = \{0, \dots, n-2\}$ after $\lambda(q_i)$ and before $\lambda(q_{i+1})$, cyclically. Then

$$g_i \geq 0, \quad \sum_{i=0}^4 g_i = n - 6. \quad (6.2)$$

The universal return R_2 changes only the common phase. By the labelled-lift consequence of Lemma 5.3, a hole slide across q_i realizes

$$(g_{i-1}, g_i) \mapsto (g_{i-1} - 1, g_i + 1) \quad (g_{i-1} > 0). \quad (6.3)$$

These directed transfers strongly connect all weak compositions of $n - 6$ into five parts. To reverse one transfer, move the same hole forward across the other four tokens; ordinary adjacent transfers then connect every pair of weak compositions. A power of R_2 aligns the remaining phase. Therefore any two section injections with the same labelled cyclic order are mutually reachable.

Now let $\lambda^* : T_0 \hookrightarrow Q$ be any cyclic-order-preserving target injection. Choose a hole $h^* \notin \text{im}(\lambda^*)$ and a power p^u with $p^u(m) = h^*$. Then $p^{-u} \circ \lambda^*$, with p^{-u} represented by a nonnegative power of the cycle, lies in the section. Reach it by the preceding paragraph and finish with p^u . The same argument starts from any cyclic-order-preserving injection, proving both (6.1) and strong connectivity.

Theorem 6.1 is stronger than the target predicate in Theorem 5.4. It describes the entire plateau-reachable lineage fiber, not only which marked pairs can occupy the kernel.

6.2 Survivor placements and frontier count

Let $F \in \mathfrak{F}_{d_n}(P_0)$. By Theorem 5.4, F is a consecutive source pair. Write the positive source order uniquely as

$$(f_0, f_1, r_1, r_2, r_3), \quad F = \{f_0, f_1\}, \quad (6.4)$$

where f_1 immediately follows f_0 .

Theorem 6.2 (canonical survivor and frontier classification).

In the notation (6.4),

$$\mathfrak{S}_{d_n}(P_0; F) = \{s : \{r_1, r_2, r_3\} \hookrightarrow \{1, \dots, n-2\} : 1 \leq s(r_1) < s(r_2) < s(r_3) \leq n-2\}. \quad (6.5)$$

Consequently,

$$|\mathfrak{B}_{d_n}(P_0)| = |\text{Fr}_{5 \rightarrow 4, d_n}^{\text{raw}}(P_0)| = 5 \binom{n-2}{3}. \quad (6.6)$$

Proof. Let λ be terminal with $\lambda(F) = K = \{0, 1\}$. Cyclic-order preservation forces

$$\lambda(f_0) = 0, \quad \lambda(f_1) = 1, \quad 2 \leq \lambda(r_1) < \lambda(r_2) < \lambda(r_3) \leq n-1. \quad (6.7)$$

The strict d_n -step sends each survivor to

$$s(r_i) = d_n(\lambda(r_i)) = \lambda(r_i) - 1,$$

giving the inequalities in (6.5).

Conversely, choose any $1 \leq u_1 < u_2 < u_3 \leq n-2$ and define

$$\lambda^*(f_0) = 0, \quad \lambda^*(f_1) = 1, \quad \lambda^*(r_i) = u_i + 1. \quad (6.8)$$

This injection preserves the source cyclic order. Theorem 6.1 supplies a plateau path to it. The next d_n -step is strict and has survivor placement $s(r_i) = u_i$. This proves (6.5).

There are five consecutive source pairs and $\binom{n-2}{3}$ increasing survivor triples for each. Different pairs have different two-element fibers over the collision root. Theorem 4.1 identifies boundary-incidence maps with raw endpoints, proving (6.6).

Equations (5.7) and (6.5) give a path-free membership test for the full canonical incidence spectrum. A map $\beta : T_0 \rightarrow \{0, \dots, n-2\}$ belongs to $\mathfrak{B}_{d_n}(P_0)$ exactly when:

1. $\beta^{-1}(0)$ is a consecutive pair in the cyclic order of T_0 ;
2. β is injective on the other three source lineages; and
3. after orienting the fused pair as in (6.4), the survivor coordinates are strictly increasing in the remaining positive source order.

No path search occurs in this criterion.

6.3 Independence from packet masses and atom composition

The dynamics above use source packets as distinguishable lineage labels. Their sizes and internal atom structure never enter the reachability proofs.

Corollary 6.3 (source-block independence of the raw incidence frontier).

Let $P_0 = (T_0, \Pi_0)$ and $P'_0 = (T'_0, \Pi'_0)$ be raw rank-five sources for the same canonical automaton, possibly on different labelled atom carriers. Let $\theta : T_0 \rightarrow T'_0$ preserve positive cyclic order. Then

$$\beta \mapsto \beta \circ \theta^{-1} \quad (6.9)$$

is a bijection

$$\mathfrak{B}_{d_n}(P_0) \xrightarrow{\sim} \mathfrak{B}_{d_n}(P'_0). \quad (6.10)$$

After replacing each source block $\Pi_0(q)$ by $\Pi'_0(\theta(q))$, formula (4.5) transports this bijection to an incidence isomorphism of the two raw first-exit frontiers. Hence the canonical raw frontier depends on source blocks only through their five lineage labels and positive cyclic order, not through packet masses or internal atom composition.

Proof. Theorem 5.4 characterizes fused pairs by cyclic adjacency, which θ preserves. For each such pair, Theorem 6.2 characterizes survivor placements solely by the induced positive order. Precomposition with θ^{-1} therefore preserves and reflects all conditions in (6.5), giving (6.10). The decoding formula (4.5) then replaces lineage labels by the corresponding packet blocks.

This is a raw incidence statement. It is not a descent theorem for typed roles, ancestry, target normalization, or any other semantics carried by a packet beyond its identity in the raw source partition.

7 Boundary to Typed Transfer and Projectability

The preceding classification answers the intrinsic first-exit question. It does not by itself answer whether an endpoint is present in a separately declared transfer relation.

Paper XXVIII fixes, for an inherited pair z , a complete transfer fiber $\mathcal{T}_{5 \rightarrow 4}(z)$ and a projectable subfiber $\mathcal{T}_{5 \rightarrow 4}^{\text{proj}}(z)$ [3]. Its open supply problem is

$$\forall z \in \text{Sec}_4^{\text{inh}}, \quad \mathcal{T}_{5 \rightarrow 4}^{\text{proj}}(z) \neq \emptyset. \quad (7.1)$$

Theorem 6.2 cannot be substituted for (7.1). The two statements read different objects:

Raw theorem provides	Typed bridge would additionally require
exact five-packet plateau and strict-exit replay	membership in the fixed complete transfer fiber
fused source pair and survivor placement	source authorization and rooted role/ancestry reads
source-addressed rank-four packet endpoint	declared target incidence and normalized typed handoff
one raw path witnessing the endpoint	a projection witness on that same transfer and entry

The first attempted promotion already reaches data absent from the raw state: authorization reads a typed source graph with rooted roles and ancestry, while projectability reads the fixed transfer binding, exact update, and handoff. This is an interface gap, not a counterexample. The raw theorem neither proves nor disproves any one of those predicates.

A future bridge may take the following conditional form. If a declared transfer relation is proved complete with respect to the canonical raw incidences of Theorem 6.2, and if one attained incidence carries the required same-witness authorization and projection data, then it supplies a projectable origin. Neither premise is established here. In particular:

$$\text{raw frontier nonempty} \Rightarrow \text{declared transfer incidence} \Rightarrow \text{projectability}. \quad (7.2)$$

No new semantic field or refined constructor is introduced to force this bridge. Projectable-Origin Supply remains motivation and future work rather than a promised conclusion of the raw classification.

8 Computational Artifacts

The paper-owned evidence package is available under [experiments/paper29/](#) in the RIME repository.

Surface	Paper-owned path	Role
Finite exact control	<code>canonical_frontier.py</code> and <code>results/</code>	Replays the intrinsic definitions for $5 \leq n \leq 12$
Partial formalization	<code>lean/</code>	Checks the data-independent incidence, quotient, counting, and boundary spine
Release validation	<code>release-manifest.json</code> and the public receipt under <code>results/</code>	Binds the manuscript, reader PDF, bibliography, development closure, and validators

The finite audit is not the proof of the all- n theorem, and the Lean development does not formalize the geometric hole-slide argument. The public receipt records local closure verification and performed replay; it is not an independent mathematical validation.

9 Claim Status and Boundary

Claim surface	Status	Scope
Faithful defect-anchored packet encoding	Proposition 3.1	Every labelled binary-kernel rank- $(n - 1)$ defect, $n \geq 5$
Coarse quotient non-descent	Proposition 3.2	Explicit five-state matched control
Permutation-plus-collapse factorization	Proposition 3.3	Every rank- $(n - 1)$ defect after choosing a kernel orientation
Lineage lift and boundary-incidence spectrum	Theorem 4.1	Every fixed raw rank-five source and labelled defect
Exact 2 + 3 spectator-safe quotient	Theorem 4.2	Fused-pair reachability for every labelled defect
Canonical fused-pair criterion	Theorem 5.4	All $n \geq 5$ for d_n
Canonical order-fiber transitivity	Theorem 6.1	All $n \geq 5$ for d_n
Complete survivor spectrum and frontier count	Theorem 6.2	All $n \geq 5$ for d_n
Source-block independence of the incidence frontier	Corollary 6.3	Sources with the same labelled cyclic-order skeleton
General-defect path-free classification	Open	Off-kernel transport may alter token order and survivor placement
Typed transfer incidence and Projectable-Origin Supply	Open bridge	Not decided by raw packet geometry alone

All positive claims above have direct mathematical proofs in the manuscript. Finite graph searches used during development are sanity checks, not premises, certificates, or an additional evidence level. The paper claims no shortest safe-word formula, path-length bound, reset threshold, credit settlement, uniform recursive menu, or all-rank descent theorem.

10 Conclusion

Rank-five first exit is neither a support-only event nor an ordinary pair-hit. The correct raw object retains source-packet lineages until the first strict defect step. For fused-pair reachability, those lineages admit an exact $S_2 \times S_3$ quotient: a marked pair moving with three spectators under a partial token/hole dynamics.

For the canonical single-defect circular family, that dynamics has a complete normal form. Guarded moves preserve the cyclic order of the five non-hole tokens, while section returns slide holes through that order. A pair reaches the kernel exactly when its source lineages are consecutive. The entire cyclic-order fiber is plateau-reachable, which then makes every increasing survivor placement attainable. The resulting raw frontier has exactly $5^{\binom{n-2}{3}}$ endpoints and is independent of packet masses and internal atom composition at the incidence level.

Two natural problems remain. For a general defect $d = \rho_d \circ e_K$, one may ask which quotient of the off-kernel transport ρ_d determines spectator-safe reachability and survivor placement. At the typed level, one may ask when the canonical raw incidences are realized inside a declared transfer fiber with a same-witness projection. The first is a generalization of the intrinsic geometry developed here. The second is the open bridge to Projectable-Origin Supply. Neither is needed for the complete canonical raw classification proved in this paper.

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